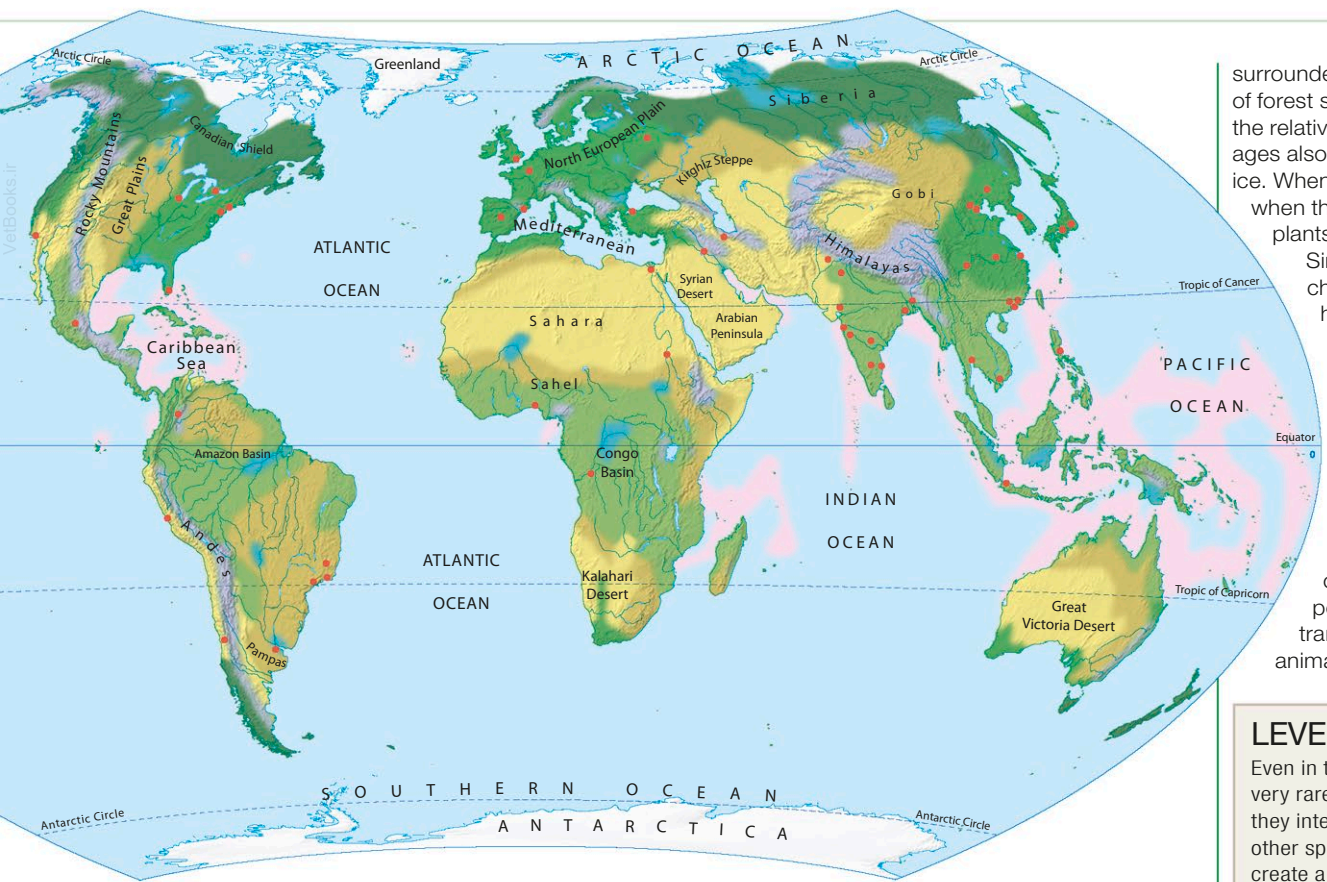




surrounded by grassland. Even today, these areas of forest still contain a wider variety of birds than the relatively new forest that has grown back. Ice ages also affect sea levels, by locking up water as ice. When sea levels fall, land habitats expand; when they rise, the land is drowned again and plants and animals are forced to retreat. Since the last ice age ended, natural changes have not been the only ones that have affected the world's habitats. On a local and global scale, human activity has had an increasing impact and, as a result, the pattern we now see is partly man-made. This is especially true of forests, which have been cut back to make space for agriculture, but it is also true of some grasslands, wetlands, and even deserts. In some remote regions—particularly in the far north—the original pattern still remains, but in populated regions, it has been transformed, creating a world where wild animals can have difficulty finding a home.

The reasons for such variation in biodiversity are still not fully understood, although climate almost certainly plays a part. However, in an age in which many animal species are endangered, biodiversity—and ways of maintaining it—has become an important topic. Tropical forests and coral reefs are especially rich in species, which is why so much attention is currently focused on preserving them and their animal life.

Animal distributions

A glance at the map on this page shows that various types of habitat are spread across large expanses of the world. However, with a few exceptions, most of their animals are not. Instead, each species has a characteristic distribution, which comes about partly through its evolutionary history (see p.17), and partly as a result of its way of life.

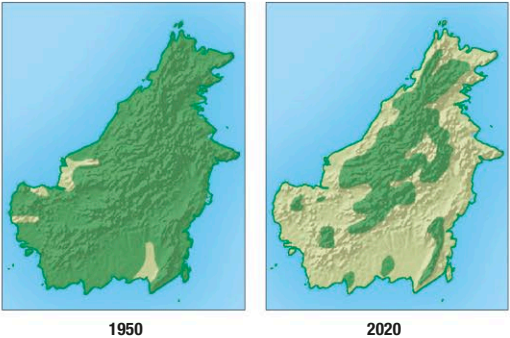
In many cases, an animal's lifestyle shapes its distribution in unexpectedly subtle ways. For example, in the Americas, the brown pelican is found all along the western coast, apart from the far north and south; in the east, it does not reach south of the Caribbean. The reason for this is that, unlike its relatives, the brown pelican feeds by

diving for fish, and needs clear water to spot its prey. The Caribbean is clear, but farther south lies the Amazon River, which pours vast amounts of mud-laden water into the sea. For the pelican, this muddy water is a barrier that cannot be crossed. Many animals' distributions are linked to those of particular plants. Extreme examples include the yucca moth, which depends on yucca plants; the fig wasp, which develops inside figs; and countless bees that depend on particular flowers. Not all plant-dependent animals are insects, however. The robber crab—the largest and heaviest terrestrial crustacean—feeds predominantly on rotting coconuts, which it scavenges along the shore. As a result, it is found only where coconuts grow. Mammals can be just as particular. The giant panda—one of the most famous examples—depends on about 2 dozen species of bamboo, which are found only at mid-altitudes in the mountains of central China.

Changing habitats

In nature, habitats change all the time. Forest and grassland catch fire, rivers burst their banks, and storms batter coral reefs and coasts. Such unpredictable occurrences are facts of life, and animals—along with other living things—have evolved ways of surviving them. Habitats can also change in much more profound ways, over much longer periods of time. Here, the driving force is usually climate change, a natural process that is triggered by a host of factors, including continental drift. On several occasions in the distant past—most recently about 12,000 years ago—the polar ice caps expanded, destroying existing habitats and evicting their animals. On each occasion, when the ice eventually melted, plants moved back into the empty landscape, and animals followed suit.

The world's climates are interrelated, which means that changes in one area can have long-term effects all over the globe. For example, during the last ice age, the climate in the tropics became drier, and the Amazon rainforest shrank to form scattered "refugia"—islands of forest



IMPOVERISHED RAINFOREST
Borneo—an island of extraordinary biological richness—had extensive rainforest 60 years ago (left). Deforestation (projected 2020 cover, right) not only endangers many species and renders others extinct, but can cause erosion and even change local climate.

LEVELS OF LIFE

Even in the remotest places on earth, animals very rarely live entirely on their own. Instead, they interact with other individuals, and with other species. Taken together, these interactions create a range of different ecological levels, from local populations, communities, and ecosystems to the whole biosphere—the sum total of all the places where living things can be found. Because microorganisms are so widespread, the biosphere extends high into the atmosphere, and probably several miles underground.



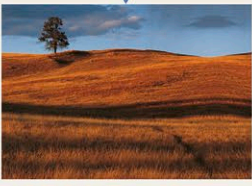
INDIVIDUAL
An individual animal is normally an independent unit that finds its own food. It often lives within a set home range.



POPULATION
A population is a group of individuals that belong to the same species, live in one area, and interbreed.



COMMUNITY
A community is a collection of populations. Although they belong to different species, they depend on each other for survival.



ECOSYSTEM
An ecosystem is made up of a community and its physical surroundings. Characteristic ecosystem types are known as biomes (or habitats).



BIOSPHERE
The biosphere consists of all the ecosystems on Earth, and therefore all the places—from the Earth's crust to the atmosphere—that living things inhabit.